

WEEK 9

Problem Statement:

Develop a User Interface for any real-world scenario using React.js (JSX — JavaScript Extension). The application must include, at minimum, a Registration Form.

Aim:

To develop a Student Registration Form using React.js, demonstrating component-based UI development, JSX syntax, the useState hook, controlled form inputs, client-side validation, and conditional rendering, styled using simple CSS.

Description:

This experiment demonstrates building a user interface using React.js for a real-world scenario — a student registration system. The application is built using Vite as the build tool and consists of a single functional component, RegistrationForm, which uses multiple useState hooks to manage each form field individually (name, email, password, confirmPassword, age, gender, terms, error, submitted). On submission, the component validates that all required fields are filled, that the password is at least 6 characters long, that the password and confirm-password fields match, and that the Terms and Conditions checkbox is agreed to — displaying an inline error message if any check fails. Once validation passes, the component conditionally renders a success screen welcoming the registered student, with an option to reset and register another student. The form is styled using a separate CSS file with a centered card layout, rounded corners, and a soft shadow.

Program:

Student Registration Form Using React.js (Vite)

Program Code :

package.json

```
{
  "name": "student-registration",
  "private": true,
  "version": "0.0.0",
  "type": "module",
  "scripts": {
    "dev": "vite",
    "build": "vite build",
    "preview": "vite preview"
  },
  "dependencies": {
    "@vitejs/plugin-react": "latest",
    "vite": "latest",
    "react": "latest",
    "react-dom": "latest"
  },
  "devDependencies": {}
}
```

vite.config.js

```
import { defineConfig } from "vite";
import react from "@vitejs/plugin-react";

export default defineConfig({
  plugins: [react()],
  server: {
    port: 5174,
  },
});
```

index.html

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Student Registration</title>
  </head>
  <body>
    <div id="root"></div>
    <script type="module" src="/src/main.jsx"></script>
  </body>
</html>
```

src/main.jsx

```
import { StrictMode } from "react";
import { createRoot } from "react-dom/client";
import "../index.css";
import App from "../App.jsx";

createRoot(document.getElementById("root")).render(
  <StrictMode>
    <App />
  </StrictMode>
);
```

src/index.css

```
body {
  margin: 0;
}
```

src/App.jsx

```
import RegistrationForm from "../RegistrationForm";

function App() {
  return <RegistrationForm />;
}

export default App;
```

src/RegistrationForm.jsx

```
import { useState } from "react";
import "../RegistrationForm.css";

function RegistrationForm() {
  const [name, setName] = useState("");
  const [email, setEmail] = useState("");
  const [password, setPassword] = useState("");
  const [confirmPassword, setConfirmPassword] = useState("");
  const [age, setAge] = useState("");
  const [gender, setGender] = useState("Male");
  const [terms, setTerms] = useState(false);
  const [error, setError] = useState("");
  const [submitted, setSubmitted] = useState(false);

  function handleSubmit(e) {
    e.preventDefault();
    setError("");

    if (!name || !email || !password || !confirmPassword) {
      setError("Please fill all required fields.");
      return;
    }

    if (password.length < 6) {
```



```

        <label>Email *</label>
        <input
            type="email"
            placeholder="student@example.com"
            value={email}
            onChange={(e) => setEmail(e.target.value)}
        />
    </div>

    <div className="form-group">
        <label>Password *</label>
        <input
            type="password"
            placeholder="Enter password"
            value={password}
            onChange={(e) => setPassword(e.target.value)}
        />
    </div>

    <div className="form-group">
        <label>Confirm Password *</label>
        <input
            type="password"
            placeholder="Confirm your password"
            value={confirmPassword}
            onChange={(e) => setConfirmPassword(e.target.value)}
        />
    </div>

    <div className="form-group">
        <label>Age</label>
        <input
            type="number"
            placeholder="Enter your age"
            value={age}
            onChange={(e) => setAge(e.target.value)}
        />
    </div>

    <div className="form-group">
        <label>Gender</label>
        <select
            value={gender}
            onChange={(e) => setGender(e.target.value)}
        >
            <option value="Male">Male</option>
            <option value="Female">Female</option>
            <option value="Other">Other</option>
        </select>
    </div>

    <div className="terms">
        <input
            type="checkbox"
            checked={terms}
            onChange={(e) => setTerms(e.target.checked)}
        />
        <span>I agree to the Terms and Conditions</span>
    </div>

    {error && <p className="error">{error}</p>}

    <button type="submit" className="register-button">
        Register
    </button>
</form>
</div>
</div>

```

```

    );
}

export default RegistrationForm;

src/RegistrationForm.css

* {
  box-sizing: border-box;
}

body {
  margin: 0;
  font-family: Arial, sans-serif;
  background: #f3f4f6;
}

.page {
  min-height: 100vh;
  display: flex;
  justify-content: center;
  align-items: center;
  padding: 30px;
}

.registration-container {
  width: 100%;
  max-width: 500px;
  background: white;
  padding: 35px;
  border-radius: 15px;
  box-shadow: 0 8px 25px rgba(0, 0, 0, 0.12);
}

.registration-container h1 {
  text-align: center;
  margin-bottom: 8px;
  color: #222;
}

.subtitle {
  text-align: center;
  color: #777;
  margin-bottom: 30px;
}

.form-group {
  margin-bottom: 18px;
}

.form-group label {
  display: block;
  margin-bottom: 7px;
  font-weight: bold;
  color: #333;
}

.form-group input,
.form-group select {
  width: 100%;
  padding: 12px;
  border: 1px solid #ccc;
  border-radius: 7px;
  font-size: 15px;
  outline: none;
}

.form-group input:focus,
.form-group select:focus {

```

```
border-color: #646cff;
box-shadow: 0 0 4px rgba(100, 108, 255, 0.3);
}

.terms {
  display: flex;
  align-items: center;
  gap: 8px;
  margin: 20px 0;
  font-size: 14px;
  color: #444;
}

.terms input {
  width: 16px;
  height: 16px;
}

.error {
  background: #ffe5e5;
  color: #d00000;
  padding: 10px;
  border-radius: 6px;
  text-align: center;
  margin-bottom: 15px;
}

.register-button,
.success-box button {
  width: 100%;
  padding: 13px;
  border: none;
  border-radius: 7px;
  background: #646cff;
  color: white;
  font-size: 16px;
  cursor: pointer;
}

.register-button:hover,
.success-box button:hover {
  background: #535bf2;
}

.success-box {
  width: 100%;
  max-width: 500px;
  background: white;
  padding: 40px;
  border-radius: 15px;
  text-align: center;
  box-shadow: 0 8px 25px rgba(0, 0, 0, 0.12);
}

.success-box h1 {
  color: #16803c;
  margin-bottom: 15px;
}

.success-box p {
  color: #555;
  margin-bottom: 15px;
}
```

Output:

1) Registration Form (initial state)

Student Registration

Create your student account

Full Name *

Email *

Password *

Confirm Password *

Age

Gender

☐ I agree to the Terms and Conditions

Register

2) After Successful Submission

Registration Successful! ✓

Welcome, **Andaluri Shanmukharao!**

Your registration has been completed successfully.

Register Another Student

Result:

The Student Registration Form was successfully developed using React.js. The application demonstrates functional components, multiple useState hooks, controlled inputs, form validation (required fields, password length, password match, terms agreement), conditional rendering of a success screen, and clean, simple CSS styling.